From Gas Physics to Cosmological Observables

SLICK connects the physical conditions of gas in cosmological simulations to the atomic and molecular line emission that telescopes observe.

01

Multi-line predictions

MOLECULAR

CO J = 1–0 … 13–12

NEUTRAL CARBON AND CARBON IONS

[C I] 609 µm

[C I] 370 µm

[C II] 158 µm

NEUTRAL OXYGEN

[O I] 63 µm

[O I] 145 µm

IONIZED GAS

[N II] 122, 205 µm

[O III] 52, 88 µm

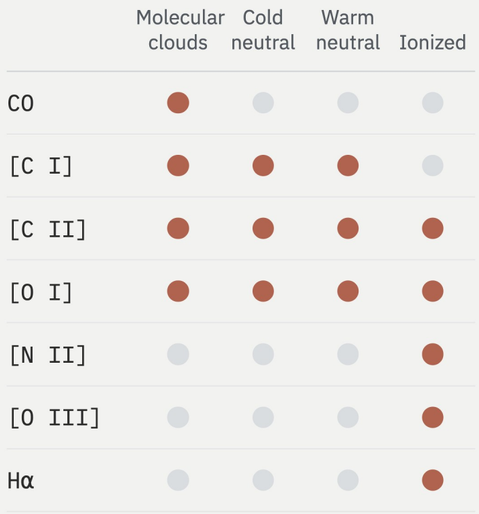
Hα, Hβ

Original SLICK SLICK II (multiphase)

Molecular and atomic tracers from the same calculation. Line labels only; no luminosities are shown.

02

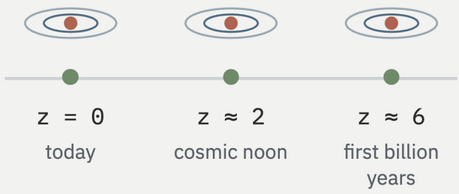
Connecting emission to gas phases



Each phase is modelled separately, so multiphase lines such as [C II] can be traced to every component that emits them.

03

Evolution across cosmic time



Line properties

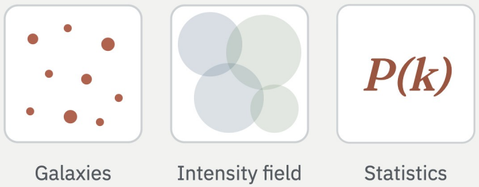
Luminosity functions

Line ratios

SLICK enables predictions of emission-line properties, luminosity functions, and line ratios across cosmic time.

04

Toward line-intensity mapping



Combining emission from galaxies across cosmological volumes enables predictions for line-intensity maps and power spectra.

Line emission is computed from the modelled physical state of the gas rather than from galaxy-wide luminosity scaling relations; sub-grid ingredients such as the cloud mass function and H II-region conditions follow empirical prescriptions.

Schematic · no quantitative results shown